Course Code: CSA0712

Course Name: Computer Networks

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NETWORK LAYER CASE STUDY REPORT

A Case Study on IPv4 Address Exhaustion in a Retail Branch

INTRODUCTION:

This case study examines an IPv4 addressing failure at a retail store branch after the deployment of 50 additional digital cash registers. The branch originally operated on the 192.168.1.0/24 subnet. During peak operating hours, devices began losing connectivity and new devices were unable to obtain an address. The DHCP server reported “no free addresses available,” providing direct evidence that the address pool had become exhausted.

The problem is a Network Layer addressing and capacity issue rather than a failure of an individual application. A /24 network contains 256 total IPv4 addresses and 254 usable host addresses. In a production retail environment, those addresses are consumed by point-of-sale terminals, employee computers, printers, access points, servers, mobile devices, surveillance equipment, and management interfaces. The theoretical capacity of 254 hosts therefore did not provide enough practical headroom for growth.

The recommended redesign is a /23 CIDR network. The proposed network is 192.168.0.0/23 with subnet mask 255.255.254.0. It contains 512 total addresses and 510 usable host addresses. This provides more than twice the usable capacity of the existing /24 and gives the branch sufficient space for the 50 new registers and future expansion.

The redesign is carefully planned to avoid unnecessary changes to static infrastructure. Existing printers can retain their current IP addresses, such as 192.168.1.20, while their subnet mask is changed from 255.255.255.0 to 255.255.254.0. The default gateway can remain 192.168.1.1. DHCP exclusions and reservations must be reviewed to prevent conflicts.

Implementation should be performed during a low-traffic period, after configuration backups and an inventory of static devices.

Problem Statement

Scenario

A retail store chain has added 50 digital cash registers to a local branch. The branch network was originally designed around the private IPv4 subnet 192.168.1.0/24. The number of connected devices has grown over time, but the addressing plan has not been expanded. The new registers create additional DHCP demand and consume the remaining free addresses.

Observed Symptoms

- New cash registers intermittently fail to obtain an IPv4 address.
- Employees report random loss of network connectivity during peak hours.
- Some devices receive no DHCP lease.
- Applications that depend on network connectivity become unavailable.
- DHCP server logs show “no free addresses available.”
- Connectivity appears to recover when devices are disconnected or leases expire.

Problem Statement

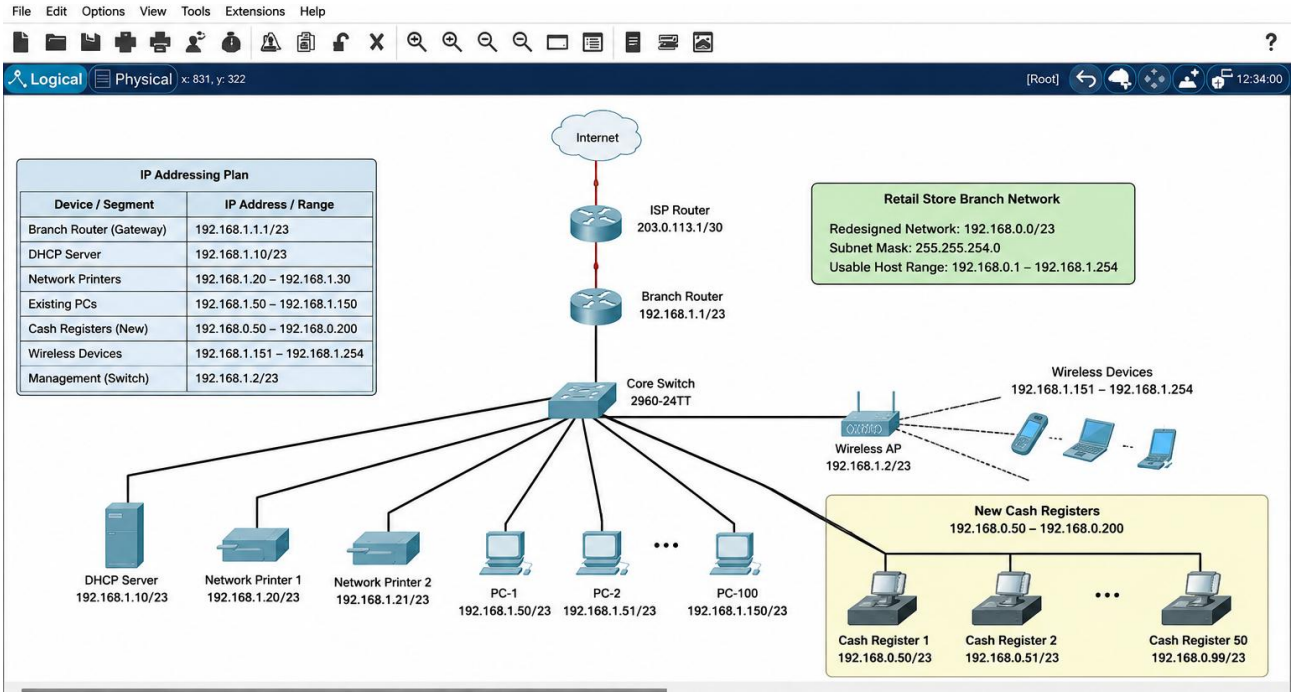
The branch has outgrown the practical capacity of its existing /24 IPv4 subnet. The immediate challenge is to increase address capacity while preserving the existing network prefix base and minimizing disruption to static devices such as printers.

Scope

The analysis focuses on IPv4 addressing, CIDR, subnet masks, DHCP capacity, static addressing, gateway configuration, and Network Layer verification. Routing protocols such as OSPF and BGP are not the primary cause of this incident because the evidence points to local address-pool exhaustion.

Network Topology and Failure Point

The following logical topology represents the affected branch. The failure point is associated with the IPv4 address allocation boundary rather than the physical switch fabric.



Original Network

Parameter	Original Configuration
Network	192.168.1.0/24
Subnet mask	255.255.255.0
Total addresses	256
Usable hosts	254
Network address	192.168.1.0
Broadcast address	192.168.1.255
Typical gateway	192.168.1.1

Failure Point

The original DHCP scope cannot allocate additional addresses once all usable addresses are consumed or reserved. This explains why devices may appear to drop randomly: devices with existing valid leases can continue to operate, while new or renewing clients may fail when they request an available address.

Technical Background: IPv4 and CIDR

IPv4 Address Structure

An IPv4 address contains 32 bits divided into a network portion and a host portion. CIDR notation specifies the number of leading bits used for the network. In 192.168.1.0/24, the first 24 bits identify the network and the final 8 bits identify hosts.

Meaning of /24

A /24 leaves eight host bits. Therefore, the address space contains 2^8 = 256 total addresses. The first address is reserved as the network address and the last is reserved as the broadcast address, leaving 254 conventional host addresses.

Meaning of /23

A /23 leaves nine host bits. Therefore, it contains 2^9 = 512 total addresses and 510 usable host addresses. The subnet mask is 255.255.254.0 because the third octet contains seven network bits and one host bit.

CIDR	Host Bits	Total Addresses	Usable Hosts
/25	7	128	126
/24	8	256	254
/23	9	512	510
/22	10	1024	1022

Why /23 Is Recommended

A /25 would provide only 126 usable hosts and is insufficient. A /24 provides 254 usable hosts but is already exhausted in the scenario. A /23 provides 510 usable hosts and therefore supplies substantial growth capacity without requiring a major architectural redesign.

The proposed /23 network must be aligned on an even boundary in the third octet. Therefore, the correct supernet containing the existing 192.168.1.0/24 is 192.168.0.0/23, not 192.168.1.0/23.

CIDR Recalculation and Address Plan

Host Capacity Calculation

For a subnet with h host bits, the number of usable addresses is 2^h - 2. For /23, h = 32 - 23 = 9.

Usable hosts = 2^9 - 2 = 512 - 2 = 510.

Thus the redesigned network provides 510 usable addresses, compared with 254 in the original network.

Address Range

Item	Value
Network	192.168.0.0/23
Subnet mask	255.255.254.0
First usable	192.168.0.1
Last usable	192.168.1.254
Broadcast	192.168.1.255
Total addresses	512
Usable host addresses	510

Addressing Plan

Device Category	Suggested Range / Method
Router / gateway	192.168.1.1
Network infrastructure	192.168.0.1–192.168.0.50
Static printers	Existing 192.168.1.x addresses retained
Servers	Reserved/static addresses
DHCP clients	Remaining addresses across /23
New cash registers	DHCP reservations or dynamic leases
Wireless clients	DHCP pool

Future devices	Remaining /23 capacity

Important Design Constraint

The existing /24 is contained inside the new /23. The new /23 combines the 192.168.0.0/24 and 192.168.1.0/24 blocks into one contiguous address space. Before deployment, administrators must verify that 192.168.0.x is not already used by another network, VPN, VLAN, or routed segment.

Evidence and Troubleshooting

DHCP Evidence

Illustrative DHCP Log

[10:14:22] DHCPDISCOVER from CashRegister-047

[10:14:22] DHCPOFFER unavailable

[10:14:22] Scope 192.168.1.0/24: no free addresses available

[10:14:25] DHCPREQUEST retry from CashRegister-047

[10:14:25] DHCPNAK / allocation failure

Client-Side Evidence

A client that cannot obtain a DHCP lease may show an automatic private address or no usable IPv4 configuration. On Windows, ipconfig /all can be used to inspect the assigned address, subnet mask, DHCP server, and default gateway.

Troubleshooting Sequence

1. Check physical link and switch-port status.
2. Check whether the client has an IPv4 address.
3. Inspect DHCP scope utilization.
4. Confirm the DHCP scope subnet and mask.
5. Review exclusions and reservations.
6. Check for duplicate or stale leases.
7. Review the number of active devices.
8. Compare actual demand with subnet capacity.
9. Calculate an expanded CIDR block.
10. Validate that the new block does not overlap another network.

Network Layer Root Cause Analysis

Root Cause

The root cause is IPv4 address exhaustion. The original /24 subnet has only 254 conventional host addresses. As the retail branch accumulated endpoints, the number of usable addresses became insufficient. The addition of 50 cash registers increased demand and caused DHCP scope exhaustion.

Why Existing Devices May Still Work

Address exhaustion does not necessarily disconnect every device simultaneously. Devices that already possess valid leases may continue using their addresses until the lease expires or the device disconnects. New devices and devices attempting to renew can fail because the DHCP server cannot allocate a free address. This produces the reported pattern of intermittent or random connectivity problems.

Network Layer Concepts

- Logical addressing: each host requires a usable IPv4 address.
- Subnetting: the mask determines which addresses belong to the local network.
- CIDR: prefix length determines the size of the address block.
- Broadcast addressing: the final address of the subnet is reserved for broadcast.
- Default gateway: traffic destined outside the local subnet is sent to the router.
- DHCP interaction: hosts dynamically obtain addresses and network parameters.

Why This Is Not Primarily OSPF or BGP

OSPF and BGP are routing protocols used to exchange reachability information. The reported DHCP message directly identifies a local address allocation problem. Even if routing were perfectly configured, a host without a valid local IPv4 address could not participate normally in the IP network. Therefore, routing protocols are secondary to the addressing problem in this case.

Protocol Behavior and Network Layer Interaction

DHCP

DHCP is responsible for dynamically assigning IPv4 addresses and associated configuration such as subnet mask, default gateway, and DNS server. The common DORA sequence consists of Discover, Offer, Request, and Acknowledgment. When the pool has no free address, the expected Offer cannot be generated for a new client.

ARP

After receiving an IPv4 address, a host may use ARP to resolve a local IPv4 address to a MAC address. During troubleshooting, duplicate IP addresses should be excluded as a possible secondary issue. A correctly expanded subnet still requires unique addresses.

ICMP

ICMP is useful for testing IP reachability. Ping uses ICMP Echo Request and Echo Reply messages. After the redesign, successful pings to the gateway and known devices provide evidence that Layer 3 connectivity is working.

NAT

NAT may be used at the branch edge to translate private addresses to a public address for Internet access. NAT is not the cause of local DHCP pool exhaustion. The private subnet can be expanded without changing the basic purpose of edge NAT, provided the router and security policies permit the new addresses.

Routing Consideration

The new /23 is a single connected local network. If the branch router advertises this network to another routing domain, the routing table or route advertisement must be updated from 192.168.1.0/24 to 192.168.0.0/23 as appropriate. Any static routes, ACLs, VPN selectors, monitoring rules, or firewall policies that explicitly reference the old /24 should be reviewed.

Proposed Solution: Implementation Plan

Phase 1 – Preparation

11. Inventory all active IPv4 addresses.
12. Identify static printers, servers, routers, access points, and other infrastructure.
13. Export or back up the router configuration.
14. Back up the DHCP configuration.
15. Record DHCP reservations and exclusions.
16. Confirm that 192.168.0.0/24 is not used elsewhere.
17. Schedule the change during a low-traffic maintenance window.

Phase 2 – Gateway Configuration

The router interface can be changed to the /23 mask while retaining the gateway IP. Example Cisco IOS syntax:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.254.0
no shutdown
exit
end
write memory
```

Phase 3 – DHCP Scope

```
ip dhcp pool RETAIL_BRANCH
network 192.168.0.0 255.255.254.0
default-router 192.168.1.1
dns-server 8.8.8.8
exit
```

The DNS address is only an example. In production, the organization's approved internal DNS server should be used.

Phase 4 – Exclusions

```
ip dhcp excluded-address 192.168.0.1 192.168.0.50
ip dhcp excluded-address 192.168.1.1 192.168.1.50
```

Static Printers and Existing Infrastructure

Preservation Strategy

The safest approach is to preserve existing static IP addresses wherever possible. A printer configured as 192.168.1.20 does not need to become 192.168.0.20. The address can remain the same because 192.168.1.20 is inside the proposed /23 network.

Setting	Before	After
Printer IP	192.168.1.20	192.168.1.20
Subnet mask	255.255.255.0	255.255.254.0
Default gateway	192.168.1.1	192.168.1.1
Addressing type	Static	Static

Why the IP Address Can Remain

A /23 mask means that both 192.168.0.x and 192.168.1.x belong to the same local network, subject to the exact host address. Therefore, a printer at 192.168.1.20 remains locally reachable by clients in the same /23.

Other Static Devices

- Update subnet masks on servers and infrastructure using static addresses.
- Retain existing IP addresses unless a documented conflict exists.
- Check monitoring systems that contain IP-based device definitions.
- Review firewall and ACL rules.
- Review printer servers and print queues.
- Review VPN and remote-management rules.
- Verify DNS records where applicable.

DHCP Reservations

For cash registers and other critical devices, DHCP reservations can provide stable addressing while keeping centralized management. Each reservation should map a device identifier, such as its MAC address, to a selected IPv4 address.

Configuration Commands and Routing Adjustments

Verify Current Interface

show ip interface brief

show running-config interface gigabitEthernet 0/0

show ip route

Verify DHCP

show ip dhcp pool

show ip dhcp binding

show ip dhcp conflict

Expected Connected Route

After configuring the interface as 192.168.1.1/23, the router should identify the connected network as 192.168.0.0/23. An illustrative routing table entry is:

C 192.168.0.0/23 is directly connected, GigabitEthernet0/0

Static Route Review

If another router has a static route pointing specifically to 192.168.1.0/24, the administrator should determine whether that route remains valid for the redesigned topology. Depending on the architecture, it may need to be replaced by or supplemented with a route for 192.168.0.0/23.

ACL and Firewall Review

Security policies frequently contain network objects such as 192.168.1.0/24. These objects should be reviewed and expanded where the policy is intended to cover the entire branch /23. Care must be taken not to broaden a security rule unnecessarily.

Deployment Procedure

Step-by-Step Deployment

18. Notify branch staff about the maintenance window.
19. Back up router, DHCP, and switch configurations.
20. Export the current DHCP lease database if supported.
21. Record all static addresses and reservations.
22. Verify that 192.168.0.0/24 is not already allocated.
23. Change the router interface mask to 255.255.254.0.

24. Change the DHCP scope to 192.168.0.0/23.
25. Review and recreate exclusions and reservations.
26. Update subnet masks on static printers and servers.
27. Renew DHCP leases on test clients.
28. Test gateway, printer, DNS, and Internet connectivity.
29. Connect several new cash registers and confirm unique leases.
30. Monitor DHCP utilization and network errors.
31. Document the final configuration.

Maintenance Window

The migration should be performed during a low-traffic period because changing a host's subnet mask can temporarily interrupt connectivity. A staged rollout is preferable: configure and test a small number of devices first, then update the remaining infrastructure.

Rollback

If the new configuration causes unexpected routing, security, or addressing problems, restore the backed-up router and DHCP configuration. Static device masks can be returned to /24 only if the old network is restored. Any newly assigned 192.168.0.x addresses should be withdrawn during rollback.

Risk Assessment and Mitigation

Risk	Likelihood	Impact	Mitigation
IP overlap with another network	Medium	High	Audit existing addressing before deployment
Incorrect DHCP scope	Medium	High	Back up and validate DHCP configuration
Static printer unreachable	Medium	Medium	Update mask while retaining IP
Duplicate IP address	Low/Medium	High	Scan and document addresses

ACL blocks new range	Medium	High	Review firewall and ACL objects
Temporary downtime	Medium	Medium	Use maintenance window
Incorrect routing advertisement	Low/Medium	High	Review routing tables and static routes
Unexpected DHCP leases	Medium	Medium	Use reservations/exclusions
Human configuration error	Medium	High	Peer review and rollback plan

Operational Risks

The largest operational risk is changing the subnet definition without updating every required static device and network policy. The /23 redesign is technically simple, but implementation must account for configuration dependencies beyond the router and DHCP server.

Business Impact

Retail operations are highly dependent on network availability because digital cash registers, payment services, inventory systems and management systems communicate over the network.

Verification and Testing

Gateway Test

ping 192.168.1.1

Successful ICMP replies confirm basic IP connectivity to the branch gateway.

Printer Test

ping 192.168.1.20

The printer should remain reachable at its original address after its subnet mask is updated.

DHCP Test

Disconnect and reconnect a test cash register. Confirm that it receives an address within the new /23 pool and that the subnet mask is 255.255.254.0.

Cross-Range Test

Test communication between a device in the 192.168.0.x range and a device in the 192.168.1.x range. Both should be considered local hosts under the /23 mask.

Traceroute

tracert 8.8.8.8

On Windows, tracert can be used to verify the path toward the Internet. The first hop should normally be the branch gateway.

Router Verification

show ip route
show ip interface brief
show ip dhcp binding

The routing table should contain the new 192.168.0.0/23 connected route, the interface should show the correct mask, and DHCP bindings should show unique leases.

Expected Results and Performance Impact

Expected Results

Test	Expected Result
Gateway ping	Successful
Static printer ping	Successful
DHCP lease	New unique address assigned
Subnet mask	255.255.254.0
Default gateway	192.168.1.1
Cross-range ping	Successful within local /23
Internet access	Successful

DHCP exhaustion message	No longer occurs under normal capacity
Routing table	192.168.0.0/23 connected route

Capacity Improvement

The usable address capacity increases from 254 to 510. This is an increase of 256 usable addresses, effectively adding another /24 block while keeping the branch on one logical /23 subnet.

The number of additional usable addresses is 510 - 254 = 256. This provides substantial headroom beyond the immediate requirement of 50 new cash registers.

Long-Term Considerations

- Monitor DHCP utilization rather than waiting for exhaustion.
- Maintain an IP address management inventory.
- Reserve infrastructure addresses consistently.
- Consider VLAN segmentation if the branch becomes significantly larger.
- Use DHCP reservations for critical endpoints when appropriate.
- Document every static address and subnet dependency.

Alternative Designs and Comparison

Option A – Keep /24

Keeping the /24 would require removing devices, reducing DHCP reservations, or redesigning the network into multiple smaller subnets. It does not solve the immediate growth problem if the branch needs more than the existing pool.

Option B – Multiple VLANs

The branch could be divided into separate VLANs, for example one VLAN for cash registers, one for employee devices, and one for infrastructure. VLAN segmentation improves security and broadcast-domain control, but it introduces inter-VLAN routing and additional administrative complexity.

Option C – /23

The /23 solution is the simplest immediate capacity expansion because it provides 510 usable addresses and can preserve the existing 192.168.1.x static addresses. It is appropriate when the branch is still small enough for a single broadcast domain.

Approach	Capacity	Complexity	Suitability
Keep /24	254	Low	Poor for current growth
/23 redesign	510	Low/Medium	Recommended
Multiple VLANs	Depends on design	Medium/High	Good for larger branches
/22 redesign	1022	Medium	May be excessive for current need

Recommendation

The /23 redesign is recommended as the immediate solution. If the branch continues rapid growth, VLAN segmentation and a formal IP address management strategy should be considered rather than repeatedly enlarging one broadcast domain.

Network Layer Resilience and Lessons Learned

Resilience

Network resilience is not limited to redundant links and routers. A resilient network also requires an addressing plan that can accommodate growth. Address exhaustion is a form of capacity failure that can become a business-impacting outage even when switches, routers, and cables are functioning correctly.

Importance of Capacity Planning

The original /24 subnet had enough theoretical capacity for 254 hosts, but the network had insufficient operational headroom. Address planning should account for current devices, reserved addresses, infrastructure, expected growth, temporary devices, and future services.

Documentation

Accurate documentation is essential. The organization should maintain a table of network blocks, gateways, DHCP pools, static assignments, reservations, VLANs, and routing dependencies. This prevents accidental overlap when the network is expanded.

Monitoring

DHCP utilization should be monitored. Alerts can be configured when the scope reaches a high utilization threshold. Monitoring can identify growth trends before users experience failures.

Security Considerations

Expanding an address range can affect security controls. Firewall rules, access-control lists, VPN configurations, network monitoring, and endpoint policies should be reviewed. A broader subnet should not automatically receive broader access to sensitive resources.

Main Lessons

- IPv4 address space is a finite operational resource.
- CIDR allows networks to be resized efficiently.
- A subnet redesign must consider address alignment.
- Static devices can often retain their IP addresses.
- DHCP exhaustion can appear as random connectivity failure.
- Configuration backups and rollback plans reduce deployment risk.

Verification Summary and Conclusion

Summary of Findings

The investigation identifies DHCP address exhaustion in the 192.168.1.0/24 network as the primary cause of the retail branch connectivity problem. The addition of 50 digital cash registers increased demand beyond the available address pool.

The proposed 192.168.0.0/23 network provides 510 usable host addresses. It includes the original 192.168.1.0/24 address space and therefore allows existing static printer IPs to remain unchanged. Static devices require their subnet mask to be updated to 255.255.254.0 so that they recognize the expanded local network.

Final Configuration

Parameter	Final Recommendation
CIDR	/23
Network	192.168.0.0/23
Subnet mask	255.255.254.0

Usable range	192.168.0.1 – 192.168.1.254
Broadcast	192.168.1.255
Usable hosts	510
Gateway	192.168.1.1
Static printer IPs	Retain existing addresses
Static printer mask	Change to 255.255.254.0

Conclusion

The CIDR redesign resolves the immediate subnet exhaustion problem while minimizing disruption to existing infrastructure. The solution demonstrates the importance of Network Layer addressing, subnet calculation, DHCP management, configuration consistency, and proactive capacity planning. Verification and monitoring should continue after deployment to ensure that the expanded address space remains sufficient as the retail branch grows.

References and Appendices

References

- RFC 791 – Internet Protocol, Internet Engineering Task Force (IETF).
- RFC 4632 – Classless Inter-domain Routing (CIDR): The Internet Address Assignment and Aggregation Plan, IETF.
- RFC 2131 – Dynamic Host Configuration Protocol, IETF.
- RFC 1918 – Address Allocation for Private Internets, IETF.
- Cisco Systems – Cisco IOS IP Addressing Services and DHCP configuration documentation.
- Computer Networking: A Top-Down Approach, Kurose and Ross.
- Computer Networks, Andrew S. Tanenbaum and David J. Wetherall.

Appendix A – Quick Calculation

Existing /24: 2^(32-24) - 2 = 254 usable hosts.

Proposed /23: 2^(32-23) - 2 = 510 usable hosts.

Additional capacity: 510 - 254 = 256 usable addresses.

Appendix B – Submission Checklist

- Title page completed
- Executive summary included
- Topology diagram included
- Problem statement and scope included
- CIDR calculations shown
- DHCP evidence discussed
- Root cause explained
- Configuration commands included
- Static printer migration explained
- Risk and rollback plan included
- Verification tests included
- Conclusion included
- References included

Appendix C – Suggested Evidence to Attach

- Screenshot of original DHCP scope showing exhaustion
- Screenshot of router interface configuration
- Screenshot of show ip route after /23 configuration
- Screenshot of DHCP bindings
- Screenshot of ipconfig on a test client
- Screenshot of successful ping to the gateway
- Screenshot of successful ping to an existing printer